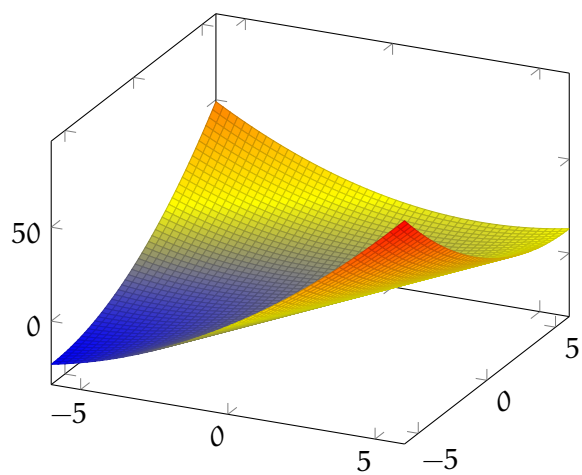


Consider the function

$$f(x_1, x_2) = \frac{1}{2}(x_1 - x_2)^2 + 3x_1 - 5,$$

which is illustrated in the figure below.



1. Write the function in quadratic form.
2. Does the function have any local minimum?